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Executive Summary

This project analyzes anonymized mobile money transaction data to classify users into behavioral activity segments.
The rebuilt dataset contains 50 real transaction users across five cohorts and 13,426 transaction rows.
The user segmentation target was defined as Low, Medium, and High activity based on user-level transaction frequency.
The best model on the refreshed build was LogisticRegression with test accuracy 1.000 and macro F1 1.000.
Behavioral intensity features such as TX_Per_Month, average amount, median amount, and send/receive balance dominate the predictive signal.
A major limitation remains: only 20 users have real demographic forms, while 30 user demographic profiles were generated to complete the analytical workflow.
Primary recommendation: use the segmentation model as an exploratory targeting tool for retention, onboarding, and product design rather than as a high-stakes decision system.

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Introduction

Mobile money usage patterns can reveal differences in transaction intensity, behavior type, and likely service needs.

The analytical objective in this project is user classification rather than amount forecasting or credit scoring.

The core research question is whether transaction history and participant demographics can separate users into low, medium, and high activity groups.

The current build uses five cohorts stored in the lab workspace: temp, lucide, rossy, precious, and jerry.

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Data Collection Methodology

The data pipeline uses five cohorts with 10 users each, giving 50 transaction participants in total.
Because anonymized user IDs were reused across cohorts, users were namespaced by cohort to avoid false merges.
Rossy required a special recovery step: its unified workbook omitted one participant, so the missing user was reintroduced from the raw P_10 message export.
Real demographic forms were available for the jerry cohort and the lucide cohort only.
This means 20 demographic profiles are collected, while 30 are generated placeholders derived from behavioral similarity.
Consent and questionnaire templates are included in the submission package.

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Exploratory Data Analysis

The transaction table contains 13,426 rows and shows a heavily right-skewed amount distribution.
Average transaction amount is 12,959.26 XAF and median amount is 1,000.00 XAF.
The EDA shows strong heterogeneity across users: some have extremely sparse monthly activity while others transact very frequently.
Grouped comparisons suggest that transaction frequency, amount profile, and direction balance are the main drivers of the eventual activity segment.
The final 50-user user-level dataset is balanced across the constructed target labels: 17 low, 16 medium, and 17 high.

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Modeling Approach

The target label Activity_Segment is derived from TX_Per_Month quantile bands.
The pipeline evaluates a baseline most-frequent classifier, Logistic Regression, Gaussian Naive Bayes, and Random Forest.
User-level train/test splitting is stratified with an 80/20 split, and model selection uses 3-fold stratified cross-validation.
The strongest test result came from LogisticRegression with accuracy 1.000, precision 1.000, recall 1.000, and macro F1 1.000.
Random Forest achieved stronger cross-validation performance than Logistic Regression, but Logistic Regression won on the held-out split.

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Results and Interpretation

The selected model separates the three activity bands cleanly on the held-out users in this run. The strongest features are listed in the feature-signal export and are dominated by activity intensity and transaction value descriptors.

num__TX_Per_Month: 0.316
num__Avg_Amount_XAF: 0.178
num__Median_Amount_XAF: 0.150
num__Send_Receive_Ratio: 0.135
num__Household size: 0.133
cat__Education level_Bachelor: 0.122
num__Total_OUT: 0.116
num__Total_Transactions: 0.115
num__Financial_Transactions: 0.110
num__Total_IN_Volume_XAF: 0.106

Example predictions are included in example_predictions.csv and show high-confidence cases for each segment class.

jerry_user007 predicted Low with confidence 0.862
temp_user0002 predicted Medium with confidence 0.921
lucide_user3333 predicted High with confidence 0.998

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Business Insights and Recommendations

High-activity users should receive premium retention and business-oriented product offers.
Low-activity users should receive onboarding and reactivation campaigns designed to increase regular usage.
Medium-activity users should be targeted with upsell and habit-building interventions to move them toward sustained high usage.
These recommendations are exploratory and should be validated against real stakeholder goals and additional field data.

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Limitations, Ethics, and Future Work

The biggest limitation is demographic incompleteness: 30 of 50 profiles are generated placeholders, not real questionnaire responses.
The target itself is constructed from the available transaction data, so it is a derived segmentation rather than an externally validated label.
The perfect held-out score for Logistic Regression should be treated cautiously because the test set still contains only 10 users.
Demographic variables can encode structural inequality, so this model should not be used for punitive or exclusionary decisions.
Future work should prioritize collecting real demographics for all cohorts, increasing the participant base, and validating the segmentation with domain experts.

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Conclusion

The five-cohort rebuild now satisfies the 50-user transaction requirement and produces an end-to-end analytical workflow from collection through modeling and interpretation.

The current package is analytically complete through Question 5 and provides the core material needed for the final report and presentation.

The main remaining risk is the partial use of generated demographics, which must be disclosed clearly in any academic or stakeholder-facing submission.